Shahriar Namvar

Chicago, IL | +1(443)440-1090 | snamva2@uic.edu | LinkedIn | GitHub | Google Scholar

SUMMARY

Ph.D. candidate in Chemical Engineering with industry experience in digital transformation, data analytics, and generative AI. Builds data-driven tools and workflows that ingests industrial or research data, automates data processing, and visualizes data for business and research problems. Skilled at explaining technical findings to diverse audiences through teaching, mentoring, and 11 peer-reviewed papers in batteries and energy storage systems.

PROFESSIONAL EXPERIENCE

LyondellBasell Industries **Houston, TX**
Digital Transformation Intern **May 2025 - December 2025**

- Built a market intelligence and sales lead tool for the Advanced Polymer Solutions (APS) business, turning Meltwater media data into a single view of competitor and OEM strategies, chemical market trends, customer sentiment, brand risks, and potential new customers.
- Designed the end-to-end workflow from automated web scraping and data pipelines to gather market and company information, keyword-based relevancy scoring to prioritize articles and prospects, and an interactive dashboard for filtering and analysis.
- Partnered with the product manager and business analysts in an agile team, gathering requirements, joining daily stand-ups, and demoing new features and UI iterations to align the tool with business needs.
- Automated the market research with an LLM workflow (LangChain, OpenAI) that screens Meltwater news, blog, and web articles for relevance to the chemical manufacturing industry and extracts contact details for sales outreach, cutting manual review effort by over 90%.
- Designed an interactive dashboard using Dash, translating Figma wireframes into a responsive UI. Enabled real-time filtering, file uploads, and visualization of both user and scraped datasets via custom callbacks.
- Wrote pytest unit tests and met production-grade development standards, with every push and pull request required to pass the team's GitHub CI checks for code formatting, quality, and engineering requirements.

Data Analyst Intern **June 2024 - January 2025**

- Developed a Python-based analytics application for predictive maintenance that integrated machine learning models, time-series data processing, and interactive user controls for model configuration and monitoring.
- Built a data processing web app to transform data from 500+ industrial equipment sources into JSON format, reducing processing time by 95% and streamlining data integration into Azure machine learning pipelines.
- Analyzed time series data from 100+ temperature sensors, identifying high-quality training and testing datasets for machine learning models, improving the overall model accuracy and predictive performance.
- Utilized Git and GitHub (Codespaces, Visual Studio) for branch development and collaborative code refinement.

RESEARCH EXPERIENCE

University of Illinois Chicago (UIC) **Chicago, IL**
Graduate Research and Teaching Assistant **June 2021 - Present**

- Designed single-metal and multivariate metal-organic framework (MOF) catalysts for nitrate electroreduction, achieving ~1 A cm^{-2} current density, 90% Faradaic efficiency, and ammonia yield rates of ~68 mg h^{-1} cm^{-2} in alkaline media.
- Synthesized and optimized advanced 3D graphene foam/MOF electrodes using chemical vapor deposition (CVD) for electrochemical energy conversion applications, achieving 1500 mA cm^{-2} current density in neutral pH conditions.
- Constructed and evaluated Zinc-nitrate batteries using galvanostatic charge/discharge testing to characterize cycling stability, power density, and degradation behavior across 50+ charge-discharge cycles.
- Developed electrochemical testing protocols for lithium-based battery systems using Biologic and Gamry potentiostat platforms with cyclic voltammetry and galvanostatic methods to evaluate reaction kinetics, catalyst stability, degradation behavior, and overall cell performance.
- Performed materials characterization and electrochemical failure analysis using XPS, SEM/EDS, Raman, XRD, and NMR to identify degradation mechanisms and performance-limiting behavior in energy storage materials.
- Developed an AI-assisted literature mining platform for MOF-based catalyst discovery by integrating Wiley TDM, CrossRef, Azure Blob Storage, and OpenAI LLMs into a scalable research workflow for accelerated materials screening.
- Published 11 peer-reviewed publications in journals including Advanced Energy Materials, Science, Advanced Materials, Nature Communications, and ACS Applied Materials & Interfaces.

SKILLS

Programming: Python, SQL, MATLAB
Data Engineering: Airflow, Snowflake, Databricks, dbt, Azure data factory, PostgreSQL
AI & Machine Learning: OpenAI APIs, LangChain, Scikit-learn
Analytics & Visualization: Dash, Plotly, Pandas, NumPy, Power BI, Tableau, DAX, Power Query
Cloud & Tools: Azure, Git, GitHub, Docker

EDUCATION

Doctor of Philosophy (Ph.D.) - Chemical Engineering **June 2021 - May 2027**
University of Illinois Chicago, Chicago, IL
Master of Science - Mechanical Engineering **June 2021 - December 2024**
University of Illinois Chicago, Chicago, IL
Bachelor of Science - Chemical Engineering **Sep 2014 - May 2019**
Amirkabir University of Technology, Tehran, Iran

PROJECTS

Daily Federal Grant Digest: An End-to-End Data Pipeline [GitHub]

- Built an end-to-end pipeline orchestrated with Apache Airflow that ingests federal grant opportunities daily from the Simpler Grants.gov API into Azure Data Lake and Snowflake, models them in dbt with a medallion architecture and SCD Type 2 history, classifies each grant by research domain with OpenAI, and publishes a daily HTML newsletter of new opportunities.

AI-Powered Research Analytics Platform for MOF Catalyst Discovery - In Progress [GitHub]

- Built an end-to-end LLM pipeline (REST API ingestion, prompt-engineered extraction, validation and analysis) that turned ~3,000 scientific publications into structured data, surfacing underexplored catalysts in various electrochemical reactions.

Generator Temperature Prediction Dashboard [GitHub]

- Built an interactive predictive maintenance dashboard in Python (Dash, Plotly) for monitoring generator temperatures. Users select sensors, operating ranges and training windows, and the dashboard retrains an XGBoost model on demand to predict temperature from current and ambient conditions, flagging sustained 24-hour deviations as early signs of overheating.

LEADERSHIP & TEACHING

- **Undergraduate Research Mentor, UIC**: Led three undergraduate students working on research project delegating tasks, guiding their experiments and data analysis, and reviewing their results.
- **Teaching Assistant/Discussion Instructor, UIC**: Led discussion sections of 40+ students and held office hours, explaining complex engineering concepts for undergraduate courses.

SELECTED PUBLICATIONS

- Intrinsically Conductive π-d Conjugated Layers with Co-N_4 Active Sites for Efficient Nitrate Electrocatalysis and Zinc-Nitrate Batteries, ACS Applied Materials and Interfaces, 2026, Link
- 2D Multivariate-Metal-Organic Frameworks for High Yield Ammonia Synthesis from Nitrate, Advanced Energy Materials, 2025, Link
- Pathways for Sustainable Reaction Kinetics in Li-CO_2 Batteries, Nature Communications, 2026, Link
- Vapor-Phase Synthesis of Electrocatalytic Covalent Organic Frameworks, Advanced Materials, 2024, Link
- Fast Charge-Transfer Rates in Li-CO_2 Batteries with a Coupled Cation-Electron Transfer Process, Advanced Energy Materials 2024, Link
- Resiliency, morphology, and entropic transformations in high-entropy oxide nanoribbons, Science, 2025, Link